Supplementary Material

Attention-Based Safe Reinforcement Learning for Scale-Agnostic and Constraint-Guaranteed V2G Charging Control

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This document supplements the main text. Section S1 proves the formal results and states the auxiliary lemmas, Section S2 documents the modelling in EV2Gym, Section S3 gives the training details, and Section S4 collects further results. Equations, lemmas and figures of the main text are cited with their main-text numbers, and items of this document carry the prefix S. Abbreviations follow the main text, which studies electric vehicle (EV) charging with vehicle-to-grid (V2G) services and photovoltaic (PV) generation. The method is the safe attention policy (SA-Attn), trained with proximal policy optimization (PPO) and compared with a multilayer perceptron (MLP), economic model predictive control (MPC) and a heterogeneous graph policy (EV-GNN). Further abbreviations are the linear program (LP), the Karush–Kuhn–Tucker (KKT) conditions and conformal prediction (CP).

S1 Proofs

This section proves the formal results of the main text and states the auxiliary lemmas that the main text cites. All statements hold for every EV separately, and the index i is dropped where no confusion arises. Inequalities are exact in real arithmetic, and the implementation checks them with one tolerance of 10^{-9} in kW or kWh that the unused part of the margin ε_q absorbs.

S1.1 Size independence of the attention policy

Lemma S1 (Equivariance and size independence)**.** *Let ϖ permute the charger tokens and their masks while the global token stays fixed.*

(i) The means μ_θ and standard deviations σ_θ are permutation-equivariant, and the value V_ω is permutation-invariant.

(ii) Outputs of unmasked tokens and the value do not depend on the features or the number of masked tokens.

(iii) The number of parameters does not depend on N.

Proof. Embeddings, layer normalisation, feed-forward blocks and the actor head act row-wise, so they commute with the permutation. For one attention head, the key mask depends only on the key index, so permuting tokens and mask maps the mask matrix M to $PMP^\top$ for the permutation matrix P that fixes the global token. Row-wise softmax satisfies $\mathrm{softmax}(PXP^\top) = P\,\mathrm{softmax}(X)P^\top$, so a masked attention head maps permuted inputs to permuted outputs, and composition gives (i) for the actor. A masked key receives zero weight in every softmax, so its value never enters the output of another token, and together with the row-wise structure this property gives (ii). Attention pooling sums over keys and is therefore invariant, and every weight matrix has a shape fixed by $(F_c, F_g, d_m, n_H, n_L)$ alone, so (iii) holds. The safety layer is permutation-equivariant as well, because its problem is invariant under joint permutation of p, x, l and u and its minimiser is unique. □

The parameter counts are 610,370 for the actor and 676,481 for the critic, 1,286,851 in total for every N. For comparison, the small ablation network ($d_m = 64$, $n_L = 2$, feed-forward width 256) has 228,067 parameters. The actor of the MLP baseline with 200 slots has 792,976 parameters, a number that grows linearly with the number of slots.

S1.2 Reachable energy and thresholds

Lemma S2 (Reachable energy)**.** *Under A1, Ψ_k is continuous and non-decreasing for every $k \geq 0$. Any admissible power sequence from $E(t)$ gives $E(t+k) \leq \Psi_k(E(t))$, with equality for full-power charging. An EV is individually feasible at step t, meaning that some admissible control meets Equation 5, if and only if $E(t) \geq B(\tau(t))$.*

Proof. A composition of continuous non-decreasing maps is continuous and non-decreasing, so $\Psi_k = \varphi^k$ inherits both properties from A1. For the reach bound we use induction on k, where the case $k = 0$ is trivial. In the induction step, an admissible power $p \leq \bar{P}(E)$ gives $E(t+1) = E + \Delta t\, g(p) \leq E + \Delta t\, g(\bar{P}(E)) = \varphi(E)$ because g is increasing, and the induction hypothesis from $E(t+1)$ with the monotonicity of Ψ_{k-1} gives

$$E(t+k) \leq \Psi_{k-1}\big(E(t+1)\big) \leq \Psi_{k-1}\big(\varphi(E)\big) = \Psi_k(E).$$

Full-power charging gives equality at every step.

For the threshold characterisation we show $\{E \in [E^{\min}, E^{\mathrm{cap}}] : \Psi_k(E) \geq E^{\mathrm{req}}\} = [B(k), E^{\mathrm{cap}}]$ by induction, where the case $k = 0$ is the definition $B(0) = E^{\mathrm{req}}$. The set in Equation 14 is non-empty because $\varphi(E^{\mathrm{cap}}) = E^{\mathrm{cap}} \geq E^{\mathrm{req}} \geq B(k-1)$. In the induction step, $\Psi_k(E) = \Psi_{k-1}(\varphi(E)) \geq E^{\mathrm{req}}$ holds if and only if $\varphi(E) \geq B(k-1)$, and since φ is continuous and non-decreasing, $\{E : \varphi(E) \geq B(k-1)\}$ is a closed up-set with minimum $B(k)$. By the reach bound, some admissible control meets Equation 5 from $E(t)$ if and only if full-power charging does, that is, if and only if $\Psi_{\tau(t)}(E(t)) \geq E^{\mathrm{req}}$. □

For a certified table, the first two properties of Equation 17 imply $\hat{B} \geq B$ componentwise, because $\varphi(\hat{B}(k)) \geq \hat{B}(k-1) \geq B(k-1)$ gives $\hat{B}(k) \geq B(k)$ by the minimality in Equation 14.

S1.3 Projection (Proposition 1 of the main text)

Proof. (i) If $p \in \mathcal{K}(t)$, then $l \leq p \leq u$ gives $l \leq u$ and $\sum_i l_i \leq \mathbf{1}^\top p \leq \sum_i u_i$, and combined with $-C_{\mathrm{rev}} \leq \mathbf{1}^\top p \leq \tilde{C}$ these inequalities give the four conditions. Conversely, $p(\upsilon) = l + \upsilon(u - l)$ stays in the box for $\upsilon \in [0,1]$, and $\mathbf{1}^\top p(\upsilon)$ takes every value in $[\sum_i l_i, \sum_i u_i]$, an interval that meets $[-C_{\mathrm{rev}}, \tilde{C}]$ under the four conditions.

(ii) The problem in Equation 19 minimises a strictly convex quadratic over a non-empty polyhedron, so its minimiser exists and is unique, and the KKT conditions are necessary and sufficient. With multipliers $\mu^+, \mu^- \geq 0$ for the aggregate constraints and $\nu^l, \nu^u \geq 0$ for the box, they read

$$p - x + (\mu^+ - \mu^-)\mathbf{1} - \nu^l + \nu^u = 0, \qquad \mu^+(\mathbf{1}^\top p - \tilde{C}) = 0, \qquad \mu^-(\mathbf{1}^\top p + C_{\mathrm{rev}}) = 0,$$
$$\nu^l_i(p_i - l_i) = 0, \qquad \nu^u_i(u_i - p_i) = 0.$$

The function $\mathcal{S}$ is continuous, non-increasing and piecewise linear, with $\mathcal{S}(\lambda) = \sum_i u_i$ for $\lambda \leq \min_i(x_i - u_i)$ and $\mathcal{S}(\lambda) = \sum_i l_i$ for $\lambda \geq \max_i(x_i - l_i)$. If $\mathcal{S}(0) > \tilde{C} \geq \sum_i l_i$, the intermediate value theorem gives a root of $\mathcal{S}(\lambda) = \tilde{C}$, and every such root is positive; the case $\mathcal{S}(0) < -C_{\mathrm{rev}} \leq \sum_i u_i$ is symmetric with a negative root. Now set $p = \mathrm{clip}(x - \lambda^*, l, u)$, $\mu^+ = \max(\lambda^*, 0)$ and $\mu^- = \max(-\lambda^*, 0)$, and choose the box multipliers coordinate by coordinate. Free coordinates with $l_i < x_i - \lambda^* < u_i$ receive $\nu^l_i = \nu^u_i = 0$, coordinates with $x_i - \lambda^* \leq l_i$ receive $\nu^l_i = l_i - x_i + \lambda^* \geq 0$ and $\nu^u_i = 0$, and the remaining ones receive $\nu^u_i = x_i - \lambda^* - u_i \geq 0$ and $\nu^l_i = 0$. Stationarity then holds coordinate-wise, and complementarity holds because $\lambda^* > 0$ only when $\mathbf{1}^\top p = \tilde{C}$ and $\lambda^* < 0$ only when $\mathbf{1}^\top p = -C_{\mathrm{rev}}$. For $\lambda^* = 0$ the point with $\mathbf{1}^\top p = \mathcal{S}(0)$ is feasible, so in all cases $p = p^*$. If $\lambda_1 < \lambda_2$ are both roots, every coordinate $\mathrm{clip}(x_i - \lambda, l_i, u_i)$ is non-increasing in λ while the sums agree, so all coordinates agree.

(iii) If the aggregate constraint is inactive, locally $p_i = x_i$ for $i \in \mathcal{F}$ and $p_i \in \{l_i, u_i\}$ is constant otherwise, so the Jacobian is D. If it is active with target $\vartheta \in \{\tilde{C}, -C_{\mathrm{rev}}\}$, then $\sum_{i \in \mathcal{F}}(x_i - \lambda) + \sum_{i \notin \mathcal{F}} p_i = \vartheta$. Hence $\partial\lambda/\partial x_j = \mathbb{1}[j \in \mathcal{F}]/|\mathcal{F}|$ and $\partial p_i/\partial x_j = \mathbb{1}[i \in \mathcal{F}]\big(\mathbb{1}[i = j] - \mathbb{1}[j \in \mathcal{F}]/|\mathcal{F}|\big)$. If $\mathcal{F} = \emptyset$, the point p is locally constant, and on the null set where p^* is not differentiable, the matrices of the adjacent regions are elements of Clarke's generalised Jacobian. □

The Jacobian is symmetric and idempotent, and as a Euclidean projection onto a convex set, Π_t is non-expansive.

S1.4 One-step invariance (Lemma 1 of the main text)

Proof. The bound $p^{\min} \le l_i$ holds by Equation 15, and for $l_i \le u_i$ note first that $p^{\min} \le 0 \le \bar{P}(E)$. Moreover, $g^{-1}\big((\hat{B}(\tau-1)-E)/\Delta t\big) \le \bar{P}(E)$ is equivalent to $\hat{B}(\tau-1) \le \varphi(E)$, and A1 with Equation 17 gives $\varphi(E) \ge \varphi(\hat{B}(\tau)) \ge \hat{B}(\tau-1)$. For the second claim, let $y = (\hat{B}(\tau-1)-E)/\Delta t$; since $l_i \ge g^{-1}(y)$ and g is increasing, every $p \ge l_i$ satisfies $g(p) \ge y$, and A3 gives $E_i(t+1) \ge E + \Delta t\, g(p) \ge \hat{B}(\tau-1)$. □

When $\tau \ge 2$, the EV is individually feasible at $t+1$ because $\tau_i(t+1) = \tau - 1$, and when $\tau = 1$, the lemma gives $E_i(d_i+1) \ge \hat{B}(0) \ge E_i^{\mathrm{req}}$.

S1.5 Feasibility invariance (Theorem 1 and Corollary 1 of the main text)

Proof of Theorem 1. We prove by induction over t the invariant I(t): $E_i(t) \ge \hat{B}_i(\tau_i(t))$ for all $i \in \mathcal{N}(t)$. An EV belongs to $\mathcal{N}(t)$ either because it arrives at t or because it was in $\mathcal{N}(t-1)$ with $d_i \ge t$. In the first case its part of the invariant is A4, in the second case the induction step applies, and since $\mathcal{N}(0)$ contains arrivals only, I(0) holds.

Assume I(t), so that every $i \in \mathcal{N}(t)$ is individually feasible and $l_i(t)$ is given by the first case of Equation 15. Lemma 1 gives $p_i^{\min} \le l_i \le u_i$, and empty chargers have $l_i = u_i = 0$. By A5 and Proposition 1(i), $\mathcal{K}(t) \ne \emptyset$, so $p(t) = \Pi_t(x(t)) \in \mathcal{K}(t)$ for every finite $x(t)$, and claim (b) follows from $p_i(t) \in [l_i, u_i] \subseteq [p_i^{\min}(E_i(t)), \bar{P}_i(E_i(t))]$. For claim (a), $\mathbf{1}^\top p(t) \in [-C_{\mathrm{rev}}(t), \tilde{C}(t)]$, and A3 gives

$$\sum_i \tilde{p}_i(t) \in \Big[\sum_i p_i(t),\ \sum_i p_i(t) + n(t)\varepsilon_q\Big] \subseteq \big[-C_{\mathrm{rev}}(t),\ \tilde{C}(t) + n(t)\varepsilon_q\big] = \big[-C_{\mathrm{rev}}(t), C(t)\big].$$

For each $i \in \mathcal{N}(t)$, Lemma 1 gives $E_i(t+1) \ge \hat{B}_i(\tau_i(t)-1)$. An EV with $d_i = t$ therefore departs with $E_i(d_i+1) \ge \hat{B}_i(0) \ge E_i^{\mathrm{req}}$, so claim (c) holds for it. Each EV with $d_i > t$ stays in $\mathcal{N}(t+1)$ with $\tau_i(t+1) = \tau_i(t) - 1$, so its part of I($t+1$) holds, and EVs that arrive at $t+1$ satisfy their part by A4. All EVs depart before the episode ends, so claim (c) holds for every EV of the episode. □

Proof of Corollary 1. The per-EV part of the induction uses only $p_i(t) \in [l_i(t), u_i(t)]$ during the stay of EV i, together with A4 and Lemma 1. Fallback outputs satisfy $p^{\min} \le p \le u$, so $p_i \ge l_i$ implies $p_i \in [l_i, u_i]$, and on feasible steps the transformer part of the induction step applies unchanged. In case (iii) of the fallback rule, Lemma S3 gives $\mathbf{1}^\top p = \tilde{C} \ge -C_{\mathrm{rev}}$, and A3 transfers this sum bound to the realised powers. □

A policy-independent sufficient condition for A5. The lower bound l_i is non-increasing in E, because both terms of its first case are, and we let $l_i^{\max}(t)$ denote l_i evaluated at $E = \hat{B}_i(\tau_i(t))$. Suppose that, at every step, $C_{\mathrm{rev}}(t) \ge 0$, $\tilde{C}(t) \ge -C_{\mathrm{rev}}(t)$ and $\sum_{i\in\mathcal{N}(t)} l_i^{\max}(t) \le \tilde{C}(t)$. Under I($t$), $l_i(t) \le l_i^{\max}(t)$ and $\sum_i u_i \ge 0 \ge -C_{\mathrm{rev}}$, so A5 holds at t, and the induction step of Theorem 1 yields I($t+1$); A5 then holds for every policy. However, $l_i^{\max}(t) = \bar{P}_i(\hat{B}_i(\tau_i))$ whenever $\hat{B}_i(\tau_i) > E_i^{\min}$, so the condition asks the transformer to serve all non-trivially constrained EVs at full power at the same time.

S1.6 Fallback rule and deficit bound

Lemma S3 (Fallback)**.** *In case (iii) of the fallback rule with $b \ge 0$, the output satisfies $p^{\min} \le p \le l$ and $\mathbf{1}^\top p = \tilde{C}$, so the transformer limit and the physical limits hold. Every EV that is individually feasible at step t and receives $p_i = l_i$ stays individually feasible. In cases (ii) and (iv), the output minimises the transformer violation over the box $[p^{\min}, u]$, so no admissible command avoids that violation.*

Lemma S4 (Deficit bound)**.** *Assume A1, A3 and A4, and assume that $\bar{P}_i$ is non-increasing in E. In the EV2Gym instantiation, the total departure shortfall of the fleet in an episode is at most the deficit of Equation 20.*

Lemma S5 in Section S2 shows that EV2Gym's taper satisfies the extra condition of Lemma S4.

Proof of Lemma S3. In case (iii), $\sum_i (l_i - p_i^{\min}) > \tilde{C} - \sum_i p_i^{\min} = b$, so the budget is exhausted before every EV reaches l_i; hence $\mathbf{1}^\top p = \tilde{C}$ and $p \le l$. Outside case (i) we also have $\tilde{C} \ge -C_{\text{rev}}$, and an individually feasible EV with $p_i = l_i$ stays feasible by Lemma 1. Over the box $[p^{\min}, u]$, the sum $\mathbf{1}^\top p$ is smallest at $p^{\min}$ and largest at u, so the outputs of cases (iv) and (ii) minimise the commanded violation. □

Proof of Lemma S4. Let $\tilde{l}_i(t) = l_i(t)$ if $E_i(t) \ge \hat{B}_i(\tau_i(t))$ and $\tilde{l}_i(t) = u_i(t)$ otherwise, and define the feasibility deficit $\Delta E^{\text{def}}(t) = (\hat{B}(\tau(t)) - E(t))^+$ with $\tau(d_i + 1) = 0$. The shortfall $(E^{\text{req}} - E(d_i + 1))^+$ is at most $\Delta E^{\text{def}}(d_i + 1)$ because $\hat{B}(0) \ge E^{\text{req}}$, and $\Delta E^{\text{def}}(t_i^{\text{arr}}) = 0$ holds by A4. It therefore suffices to show $\Delta E^{\text{def}}(t+1) \le \Delta E^{\text{def}}(t) + \Delta t \left(g(\tilde{l}) - g(p)\right)^+$ at every step and to sum these increments over the stay. If $E \ge \hat{B}(\tau)$, then $E + \Delta t\, g(l) \ge \hat{B}(\tau - 1)$ by Equation 15, and A3 gives $E(t+1) \ge \hat{B}(\tau - 1) - \Delta t\,(g(l) - g(p))^+$. When $E < \hat{B}(\tau)$ instead, $\tilde{l} = u$ and $E + \Delta t\, g(u) = \varphi(E)$, and Equation 17 with the 1-Lipschitz property of φ gives $\hat{B}(\tau - 1) - \varphi(E) \le \varphi(\hat{B}(\tau)) - \varphi(E) \le \hat{B}(\tau) - E = \Delta E^{\text{def}}(t)$. Hence $\hat{B}(\tau - 1) - E(t+1) \le \Delta E^{\text{def}}(t) + \Delta t\,(g(u) - g(p))$ in this case as well. For the fleet bound with g the identity, $p_i \ge \tilde{l}_i$ holds on feasible steps and $p = u$ holds in case (ii) of the fallback rule. In case (iii), $p \le l$ and $\sum_i (l_i - p_i) = \sum_i l_i - \tilde{C}$, and in case (iv), $p = p^{\min}$, so summing the per-step increments over all EVs gives Equation 20. □

S1.7 Conformal departure guarantee (Proposition 2 of the main text)

Proof. (i) By exchangeability, the rank of the test score $S_{n+1} = \hat{q}(\mathbf{w}_i) - d_i$ among the scores $S_1, \ldots, S_{n+1}$ is uniform up to ties, so $\mathbb{P}(S_{n+1} \le \hat{Q}_{1-\alpha}) \ge k/(n_{\text{cal}} + 1) \ge 1 - \alpha$ [1, 2]. The event $S_{n+1} \le \hat{Q}_{1-\alpha}$ is equivalent to $d_i \ge \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha}$, and because d_i is an integer, also to $d_i \ge \lceil \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha} \rceil$. Taking the maximum with t_i^{arr} changes nothing, because $d_i \ge t_i^{\text{arr}}$ holds surely.

(ii) Consider the event that $d_i \ge \hat{d}_i^{\text{lo}}$ and that $p_i(t) \ge l_i(t)$ at every step of the stay. On this event, Lemma 1 with τ^{lo}, applied up to $t = \hat{d}_i^{\text{lo}}$, gives $E_i(\hat{d}_i^{\text{lo}} + 1) \ge \hat{B}(0)$. For the remaining steps $\hat{d}_i^{\text{lo}} < t \le d_i$ we have $\tau^{\text{lo}} = 1$, and the third property of Equation 17 gives $\hat{B}(1) \le \max\{\hat{B}(0), E^{\min}\} \le E_i(t)$. Lemma 1 with $\tau = 1$ then gives $E_i(t+1) \ge \hat{B}(0) \ge E^{\text{req}}$ again, so $E_i(d_i + 1) \ge E^{\text{req}}$ holds on the event, and the union bound gives the stated probability. If A5 holds almost surely, the second probability vanishes and the bound becomes $1 - \alpha$. Note that the bound does not hold conditionally on the event that A5 held, because that event may be correlated with d_i. □

S1.8 Upper bound of the perfect-information LP

Proposition S1 (Upper bound)**.** *(a) Let π be any controller whose closed-loop trajectory satisfies Equations 2–3, satisfies Equation 6 at every step and has a total shortfall $\sum_i (E_i^{\text{req}} - E_i(d_i + 1))^+ \le z^*$. Then $J^\pi \le J^{\text{LP}}$.*

(b) In EV2Gym, let $J^{\text{LP,ceil}}$ be the optimum of the same LP with E_i^{req} and $E_i^{\min}$ lowered by $G_i^{\max} = 0.01\,(d_i - t_i^{\text{arr}} + 1)$ kWh. Then $J^\pi \le J^{\text{LP,ceil}}$ for every controller whose reported powers satisfy Equation 6 and whose realised total shortfall is at most z^*.

Proof. (a) Set $p_i^{\pm}(t) = p_i(t)^{\pm}$, take $E_i(t)$ from the trajectory and set $z_i = (E_i^{\text{req}} - E_i(d_i + 1))^+$. With these values, the energy constraints of Equation 8 hold by Equation 2 and Equation 3, and the power bounds hold because $p_i \le \bar{P}_i(E_i) \le \bar{P}_i^{\text{nom}}$ and $p_i \ge p_i^{\min} \ge -P_i^{\text{dis}}$. In addition, the departure constraint holds by the definition of z_i, and the transformer constraint is Equation 6 itself. Since $\sum_i z_i \le z^*$, the trajectory is feasible for stage 2, and its objective equals J^π.

(b) Let $\tilde{p}_i(t)$ be the reported powers and $E_i^{\text{nom}}(t) = E_i^{\text{arr}} + \Delta t \sum_{s<t} \tilde{p}_i(s)$ the energy that the billed powers alone would produce. EV2Gym bills exactly $\tilde{p}$, so J^π is the LP objective at $\tilde{p}$. The rounding adds less than 0.01 kWh per active step, so $E_i^{\text{real}}(t) - G_i^{\max} \le E_i^{\text{nom}}(t) \le E_i^{\text{real}}(t) \le E_i^{\text{cap}}$ holds along the trajectory. Both the lowered energy floor and the lowered departure constraint therefore hold for E^{nom} with $z_i = (E_i^{\text{req}} - E_i^{\text{real}}(d_i + 1))^+$, and the power bounds hold because $\tilde{p}_i \le \bar{P}_i(E_i^{\text{real}}) \le \bar{P}_i^{\text{nom}}$. From here on, the argument is as in (a). □

The main text uses $J^{\text{LP,ceil}}$ as the reference of the optimality gap, because a feasible controller inside EV2Gym can exceed J^{LP} through the unbilled rounding gain. A gap with respect to J^{LP} could then be

negative, which would be meaningless for a bound. Table S9 in Section S4 reports the gap with respect to both optima.

S2 EV2Gym modelling notes

This section documents how the model of the main text maps onto EV2Gym at commit `55ef368c`, and the simulator is used without changes to its source. All deviations from the defaults of the simulator are made through our own wrapper, configuration files and EV specification file.

S2.1 Charging taper and Assumptions A1 and A3

In state-of-charge units $\varsigma = E/E^{\mathrm{cap}}$, let $\delta = \bar{P}^{\mathrm{nom}}\Delta t/E^{\mathrm{cap}}$ be the gain of one full-power step, $\varsigma^{\mathrm{tr}} \in (0,1]$ the transition state of charge, m the taper multiplier and $\kappa = m\delta/(1-\varsigma^{\mathrm{tr}})$. When the EV's maximum alternating-current charging power does not exceed the charger rating, reading the charging routine of EV2Gym at full current gives

$$\frac{\varphi(E)}{E^{\mathrm{cap}}} = \min\{\hat{\varphi}(\varsigma), \varsigma+\delta\}, \qquad \hat{\varphi}(\varsigma) = \begin{cases} \varsigma+\delta, & \varsigma+\delta \le \varsigma^{\mathrm{tr}}, \\ 1-(1-\varsigma^{\mathrm{tr}})\exp\Big(-m\dfrac{\varsigma+\delta-\varsigma^{\mathrm{tr}}}{1-\varsigma^{\mathrm{tr}}}\Big), & \varsigma < \varsigma^{\mathrm{tr}} < \varsigma+\delta, \\ 1-(1-\varsigma)\,e^{-\kappa}, & \varsigma \ge \varsigma^{\mathrm{tr}}. \end{cases} \tag{S1}$$

Lemma S5 (A1 in EV2Gym)**.** *Suppose that the maximum alternating-current charging power of the EV does not exceed the charger rating. Then, for every step length, the full-power one-step map of EV2Gym in Equation S1 is continuous, non-decreasing and 1-Lipschitz.*

Proof of Lemma S5. At $\varsigma = \varsigma^{\mathrm{tr}} - \delta$ the first two branches of $\hat{\varphi}$ both equal ς^{tr}, and at $\varsigma = \varsigma^{\mathrm{tr}}$ the last two branches both equal $1-(1-\varsigma^{\mathrm{tr}})e^{-\kappa}$. The map is therefore continuous, as a minimum of continuous functions. Its branch slopes are 1, $m\exp(\cdot) > 0$ and $e^{-\kappa} \in (0,1)$, and $\varsigma+\delta$ has slope 1, so the map is non-decreasing as a minimum of non-decreasing functions, and A1 holds. Only the middle branch can have a slope above 1, namely when $m > 1$. On the interval $(\varsigma^{\mathrm{tr}}-\delta, \varsigma^{\mathrm{tr}})$, the function $h(\varsigma) = \hat{\varphi}(\varsigma) - \varsigma - \delta$ is strictly concave with $h(\varsigma^{\mathrm{tr}}-\delta) = 0$. If $h(\varsigma_1) < 0$ at some interior point, concavity gives $h'(\varsigma_1) \le h(\varsigma_1)/(\varsigma_1 - \varsigma^{\mathrm{tr}} + \delta) < 0$, so $\hat{\varphi}'(\varsigma_1) < 1$. Wherever the minimum in Equation S1 selects $\hat{\varphi}$ strictly, its slope is thus below 1, and elsewhere the minimum selects $\varsigma+\delta$ with slope 1. The map is therefore piecewise continuously differentiable with slopes in $[0,1]$ and hence 1-Lipschitz. □

In EV2Gym, $\eta^{\mathrm{c}} = 1$ and $\varphi(E) = E + \Delta t\,\bar{P}(E)$, so a 1-Lipschitz φ is equivalent to a non-increasing $\bar{P}$, the extra condition of Lemma S4.

Lemma S6 (A3 in EV2Gym)**.** *Suppose that, in addition to the condition of Lemma S5, efficiencies are constant over current levels and chargers and EVs are three-phase with one port and zero dead-bands. Assume further that V2G is enabled and that all EVs leave before the episode ends, and let $z_0(m) \in (0,1)$ be the positive root of $1-e^{-mz} = z$ for $m > 1$. If $\kappa \le m\,z_0(m)$ for every EV, the actuation map of Algorithm 1 realises A3 with $\varepsilon_q = 0.008$ kW.*

Proof sketch. Let $\hat{\delta} \le \delta$ be the commanded gain of a sub-maximal charging command, for which EV2Gym shifts the transition point to $\varsigma_x = 1-(\hat{\delta}/\delta)(1-\varsigma^{\mathrm{tr}})$. If the step stays below ς_x or starts above it, a direct comparison with Equation S1 shows that the delivered gain equals $\hat{\delta}$. When the step crosses ς_x, let $\Omega = 1-\varsigma_x$ and $z = (\varsigma+\hat{\delta}-\varsigma_x)/\Omega$, so that the delivered gain minus $\hat{\delta}$ equals $\Omega(1-e^{-mz}-z)$. This expression is concave in z and vanishes at 0 and $z_0(m)$, so it is non-negative for $z \le z_0(m)$. Bounding z through $\hat{\delta} \le \varphi(E)/E^{\mathrm{cap}} - \varsigma$ shows that $\kappa \le mz_0(m)$ suffices in both sub-cases $\varsigma < \varsigma^{\mathrm{tr}}$ and $\varsigma \ge \varsigma^{\mathrm{tr}}$. Discharge commands have no taper and are delivered exactly within the limits of Equation 3. Upward rounding to the five-decimal action grid adds at most 2.22×10^{-4} kW, and the discharge dead-band of 0.01 A adds at most 0.0069 kW. Sending a command within 10^{-3} kW of u_i as full current adds at most 10^{-3} kW, so the realised power lies in $[p_i, p_i+\varepsilon_q]$. Rounding the stored energy up to 0.01 kWh only adds energy, and all capacities are multiples of 0.01 kWh. □

Table S1: Scenario parameters. Quantities marked "linear" scale with the number of chargers N.

Item	Setting	Scaling with N
Simulator	EV2Gym, commit `55ef368c`, unmodified	–
Station sizes	train $N \in \{10, 20\}$; test $N \in \{5, 10, 20, 50, 100, 200\}$	–
Ports	one per charger	–
Transformer rating	$P_{\mathrm{tr}} = \rho_{\mathrm{tr}} N$, $\rho_{\mathrm{tr}} \in \{4, 7\}$ kW (tight, moderate)	linear
Inflexible load	mean peak $0.4 P_{\mathrm{tr}}$, redrawn per episode	linear
PV generation	mean peak $0.4 P_{\mathrm{tr}}$, peak below P_{tr}	linear
Demand response	off	–
Step, horizon	$\Delta t = 15$ min, $T = 112$ (28 h); sensitivity 5 min, $T = 336$	–
Profiles (start, spawn multiplier)	workplace (05:00, 5, weekdays); public (12:00, 4); private (12:00, 2.5)	per charger
Mean occupancy	about 0.21 for all profiles	–
EV models	8 V2G-capable models, constant efficiency per model	–
Taper multiplier	5 (EV2Gym default)	–
Desired energy	full battery; sensitivity 80% state of charge	–
Prices	Dutch day-ahead prices, discharge price = charge price	–
Forecast noise	Gaussian, standard deviation 10% of the value	–
Test episodes	100 per configuration ($N \leq 50$), 50 ($N \geq 100$); 3,100 per policy	–

With $m = 5$, the root is $z_0(5) = 0.99302$ and the threshold is $m z_0(m) = 4.965$. In our configuration $\kappa \leq 2.391$ at $\Delta t = 15$ min and $\kappa \leq 0.797$ at $\Delta t = 5$ min for all eight EV models, so Lemma S6 applies. For $m \leq 1$ the delivery is not exact near the transition point, and the taper multiplier therefore keeps its default value of 5. [TODO: State the agreement between the closed form of Equation S1 and EV2Gym's charging routine, taken from the unit tests of the code repository].

S2.2 Actuation, rounding and accounting

The decision variable p_i is the power that EV2Gym reports, bills and sends to the transformer, and it equals the battery-side energy change per step. With this convention $\eta_i^{\mathrm{c}} = \eta_i^{\mathrm{d}} = 1$ in Equation 2, and EV2Gym's charge and discharge efficiencies, one constant value per EV model in our specification file, act only as derating factors on the power limits. The actuation map sends a command as full current when it lies within 10^{-3} kW of u_i. Otherwise, it rounds the command upwards to the five-decimal action grid of the simulator. EV2Gym ignores discharge currents below 0.01 A and rounds the stored energy up to 0.01 kWh after every active step, so the forward limit is tightened by $n(t)\varepsilon_q$ with $\varepsilon_q = 0.008$ kW. Without this margin, realised powers would exceed the forward limit by up to about $n(t) \cdot 0.007$ kW on tight steps.

S2.3 Scenarios and arrival profiles

As seen in Table S1, every exogenous quantity scales linearly with the number of chargers N. The three arrival profiles of EV2Gym differ in their arrival times, stays and requested energies. Workplace arrivals concentrate on weekday mornings, public arrivals spread over the day with an afternoon and evening peak, and private arrivals occur in the late afternoon and evening with overnight stays. EV2Gym does not generate an EV whose departure would fall within four steps of the episode end, so overnight sessions are silently missing when the episode starts too late.

The public and private profiles therefore start at 12:00 instead of 05:00, and the share of dropped spawn attempts, measured on the arrival model alone, falls to 6.8% and 3.7%. Spawn multipliers per profile equalise the mean charger occupancy at about 0.21, and they also equalise the energy demand per charger to within about 5%. The episode starts empty, and loads and PV are redrawn for every episode. Our literature configuration reproduces EV2Gym's 25-charger V2G profit-maximisation case with 100 kW, the workplace profile and no load or PV, using our EV specification file. Figure S5 shows how training, test replays and the oracle share one scenario generator.

S2.4 Admission control

When EV j arrives at step t_0, admission control solves a feasibility LP over all connected EVs i and steps $t = t_0, \dots, d_i$, starting from the current energies:

$$\begin{aligned} & E_i(t+1) = E_i(t) + \Delta t\, p_i(t), \qquad E_i^{\min} \le E_i(t+1) \le E_i^{\mathrm{cap}}, \qquad E_i(d_i+1) \ge E_i^{\mathrm{tgt}}, \\ & -P_i^{\mathrm{dis}} \le p_i(t) \le \bar{P}_i^{\mathrm{nom}}, \qquad p_i(t) \le \rho_i \left(E_i^{\mathrm{cap}} - E_i(t)\right) \ (t > t_0), \qquad p_i(t_0) \in [p_i^{\min}, \bar{P}_i], \\ & -\hat{C}_{\mathrm{rev}}(t) \le \textstyle\sum_i p_i(t) \le \hat{C}(t) - n(t_0)\varepsilon_q. \end{aligned} \tag{S2}$$

The limits $\hat{C}$ and $\hat{C}_{\mathrm{rev}}$ use the measured values at t_0, the noisy forecasts for the next H_f steps and the last forecast afterwards. Its slope ρ_i is the largest value with $\bar{P}_i(E) \ge \min\{\bar{P}_i^{\mathrm{nom}}, \rho_i(E_i^{\mathrm{cap}} - E)\}$ for all E, so the linear taper model is an inner approximation and every LP-feasible schedule is physically realisable. Inside the LP, the new EV has the target $E_j^{\mathrm{tgt}} = E_j^{\mathrm{req}}$, and a connected EV has the target $\min\{E_i^{\mathrm{req}}, \Psi_i^{\mathrm{LP}}(E_i(t_0)) - 10^{-6}\}$. Here Ψ_i^{LP} is the largest departure energy reachable in the LP's taper model, and this target is used only inside the admission LP. With full-battery targets and the literal targets E_i^{req}, the LP would be infeasible whenever such an EV is connected, because the linear taper model never reaches full charge in finitely many steps. Feasibility is monotone in the new request, so one LP that maximises $E_j(d_j+1)$ gives the largest admissible request, and the reduced request is re-certified by the cap routine of Equation 4.

S2.5 Baseline implementation

The penalty-based learned methods use the reward $r(t)$ minus $w_E = 10\,€$ per kWh of departure shortfall and $w_P = 10\,€$ per kWh of transformer overload energy, computed with the untightened limits and the realised powers. They run without admission control, and their actor loss masks only tokens whose action cannot influence the executed power. A dense variant adds $w_E \Delta t \sum_i (l_i - p_i)^+$ to every step and serves as an ablation, because it gives the penalty method the information of our lower bound. The economic MPC of EV2Gym runs through an adapter subclass with the published formulation, and the adapter fixes two issues with newer numerical libraries without modifying EV2Gym. Beyond 25 chargers, a related HiGHS port solves a net-power LP with per-EV limits, exact first-step bounds including the taper and an elastic departure constraint, and it is reported under its own name. EV-GNN runs unmodified at commit `01aa1b7a` through a launcher that restores numerical-library aliases, with one model per arrival profile.

S3 Training details

S3.1 Training loop

Algorithm S1 lists the training loop of SA-Attn, in which the safety layer of Algorithm 1 acts as part of the environment. Each environment writes into its own buffer, and at update time all token tensors are right-padded to the largest N in the batch with masked entries. Padding changes neither the actor outputs of real tokens nor the value (Lemma S1), and the log-probability, entropy and regulariser sums run over masks only. The learning reward is the profit divided by $N r^{\mathrm{ref}}$, so that returns of stations with different N have the same scale, and every reported result uses the undivided profit.

S3.2 Hyperparameters and tuning

Table S2 lists the hyperparameters of SA-Attn and of the learned baselines. Tuning uses the tight 10-charger station and 20 validation episodes that cycle through the three profiles, with seeds disjoint from training and test seeds. For each of the four learned methods of our own code, a full grid of 18 configurations is trained for 0.5×10^6 steps with seed 0, and selection uses the training objective of the method. The conformal variant and the ablations of SA-Attn inherit the selection of SA-Attn, the penalty ablations inherit that of Attention+penalty, and EV-GNN keeps its published hyperparameters. Penalty weights, β, the architecture and all other PPO constants are fixed a priori and not tuned.

Algorithm S1 PPO training of SA-Attn with a variable number of chargers

Require: 16 environments (8 with $N = 10$, 8 with $N = 20$); actor π_θ and critic V_ω

while fewer than 3×10^6 environment steps **do**
  for 128 steps in every environment **do**
    observe tokens, l, u, $\tilde{C}$, C_{rev} and the feasibility flag
    sample $a \sim \pi_\theta(\cdot \mid s)$; $x \leftarrow P^{\text{ref}} a$
    $p \leftarrow \Pi_t(x)$ if feasible, else fallback rule ▷ Algorithm 1
    step the simulator; $r \leftarrow$ Equation 10; store $r/(Nr^{\text{ref}})$, a, $\log \pi_\theta(a \mid s)$ over tokens with $u_i - l_i > 10^{-9}$
  end for
  compute advantages with GAE ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$); pad tokens to the largest N in the batch
  for 10 epochs, minibatches of B_{mb} transitions **do**
    clipped surrogate ($\epsilon_{\text{clip}} = 0.2$) on feasible transitions, normalised by B_{mb}
    loss $\leftarrow -$surrogate $+ c_{\text{vf}}\cdot$value loss $- c_{\text{ent}}\cdot$entropy $+ \mathcal{L}_{\text{reg}}$ (Equation 22)
    Adam step with gradient-norm clipping 0.5; stop the epochs early if the approximate KL divergence exceeds 0.03
  end for
end while

Table S2: Hyperparameters and tuning grid of SA-Attn and of the learned baselines.

Group	Parameter	Value
Network	embedding width d_m, layers n_L, heads n_H	128, 3 (pre-LayerNorm), 4
	feed-forward width, dropout	512 (GELU), 0
	actor head (shared)	$128 \to 64 \to 2$
	critic readout	attention pooling, 1 query, $128 \to 64 \to 1$
	parameters (actor / critic)	610,370 / 676,481
PPO	optimiser, learning rate	Adam, tuned in $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$, linear decay
	minibatch, entropy coefficient	tuned in $\{256, 1024\}$ and $\{0, 10^{-3}, 10^{-2}\}$
	discount γ, GAE λ_{GAE}	0.99 ($0.99^{1/3}$ at 5 min), 0.95
	clip range, value coefficient	0.2, 0.5
	parallel environments, rollout length	16 (8 with $N = 10$, 8 with $N = 20$), 128
	epochs, target KL, gradient-norm clip	10, 0.03, 0.5
	environment steps per run, seeds	3×10^6, 5
Safety layer	P^{ref}, β	11 kW, 0.01
	ε_q, numerical tolerance	0.008 kW, 10^{-9}
Features	E^{ref}, c^{ref}, H_f	100 kWh, 0.25 €/kWh, 8 steps
Penalty baselines	w_E, w_P	10 €/kWh each (fixed)
Conformal	n_{cal}, α	5,000 per configuration, $\{0.01, 0.05, 0.1, 0.2\}$

S3.3 Compute budget

The main experiments train five methods with five seeds each, namely MLP+penalty, MLP+safety layer, Attention+penalty, SA-Attn and SA-Attn-CP at $\alpha = 0.1$. Four ablations with five seeds and two penalty-weight sensitivities with three seeds add 26 runs, for 51 runs of $3{\times}10^6$ steps, or about $1.5{\times}10^8$ environment steps. Tuning adds 72 short runs with 3.6×10^7 steps in total, and the optional EV-GNN baseline and the 5-min sensitivity add 15 and 5 runs. The conformal variant is evaluated at all four values of α without retraining, because Proposition 2 holds for any policy. [RESULT: measured wall-clock training and evaluation time on the hardware of the main text].

S4 Additional results

[TODO: Skeleton: fill the result placeholders from the analysis outputs, use the past tense and write the interpretation yourself].

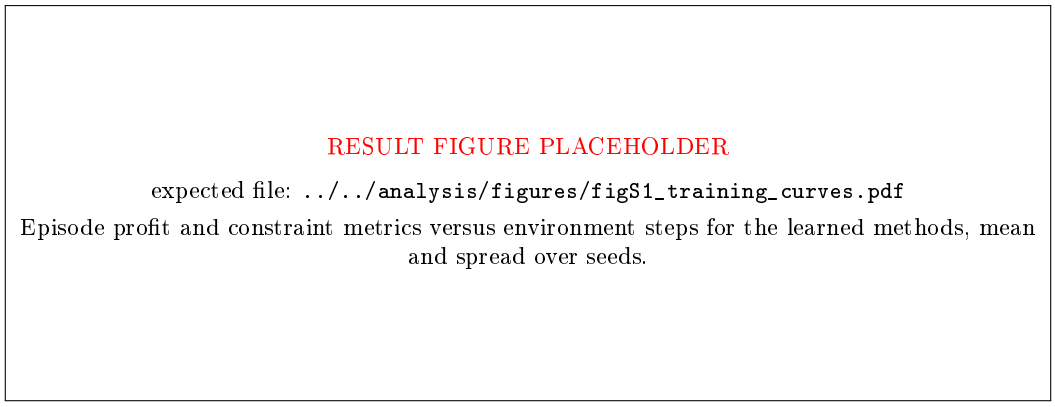


Figure S1: Training curves of the learned methods.

Table S3: Zero-shot transfer across station sizes N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

S4.1 Learning curves and full result tables

Figure S1 shows the training curves of the learned methods, and Table S3 lists the zero-shot results behind Figure 3 for every station size. Table S4 gives the constraint-satisfaction metrics behind Figure 4, and Table S5 gives the cost of the guarantee behind Figure 5. Across seeds, the training curves of SA-Attn [RESULT: description of convergence and seed spread].

S4.2 Ablation study

Figure S2 and Table S6 compare attention and MLP policies with and without the safety layer, together with the further ablations of SA-Attn. Without admission control, infeasible steps and shortfall change by [RESULT: change at the training sizes and at $N = 200$]. Setting $\beta = 0$ changes the projection distance by [RESULT: value] and the profit by [RESULT: value], and the curriculum and the small network give [RESULT: gaps relative to the full method].

S4.3 Conformal departure bound

Figure S3 and Table S7 show coverage, service and profit of SA-Attn-CP against the miscoverage level α. The episode-averaged coverage is [RESULT: coverage for each α] against the nominal levels $1 - \alpha$. Compared with known departure steps, the profit falls by [RESULT: profit loss for each α], whereas the uncalibrated point prediction gives [RESULT: coverage and shortfall of the point prediction].

S4.4 Computation time

Figure S4 and Table S8 report the inference time per control step of the policy and the safety layer, together with the training time per run. At $N = 10$ and $N = 200$, one step takes [RESULT: mean inference time in ms], of which the safety layer takes [RESULT: share]. Arrival steps with admission control take [RESULT: time of the admission LP at $N = 200$], and one oracle LP takes [RESULT: solve time].

S4.5 Optimality gap with respect to both optima and sensitivity studies

Table S9 reports the optimality gap with respect to $J^{\mathrm{LP,ceil}}$ and to J^{LP} for every station size, and the difference between the two optima is [RESULT: mean difference $J^{\mathrm{LP,ceil}} - J^{\mathrm{LP}}$ per scenario]. In the sensitivity studies of Table S10, a desired state of charge of 0.8 gives [RESULT: gap and shortfall], and a step of 5 min gives [RESULT: gap and shortfall].

Table S4: Constraint satisfaction per method over all test-grid configurations (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S5: Cost of the guarantee per station size N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

S4.6 Pipeline and example episode

Figure S5 summarises the experimental pipeline, and Figure S6 traces one test episode of SA-Attn with the aggregate power against the forward and reverse limits, the band between $\sum_i l_i$ and $\sum_i u_i$, and the price. In this episode, SA-Attn [RESULT: description of how charging and V2G discharging follow the price and the limits].

References

[1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer International Publishing, 2022.

[2] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, pp. 1094–1111, June 2018.

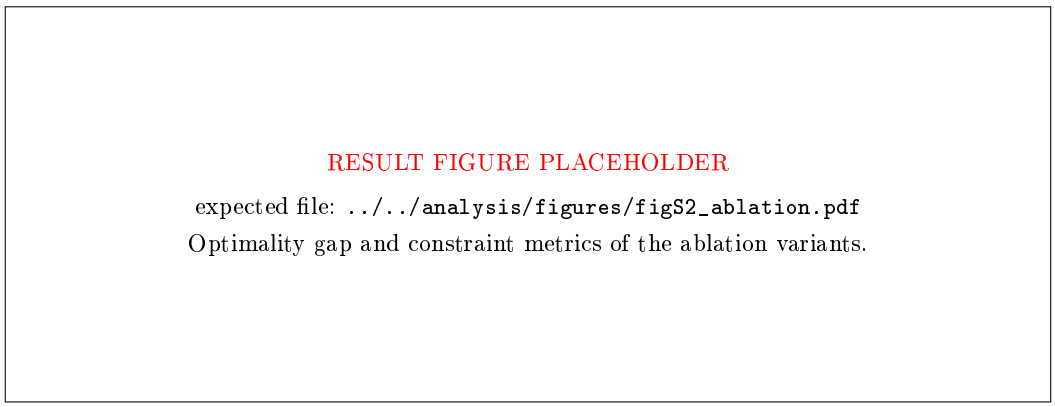


Figure S2: Ablation study.

Table S6: Ablation of architecture, constraint handling and training choices (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

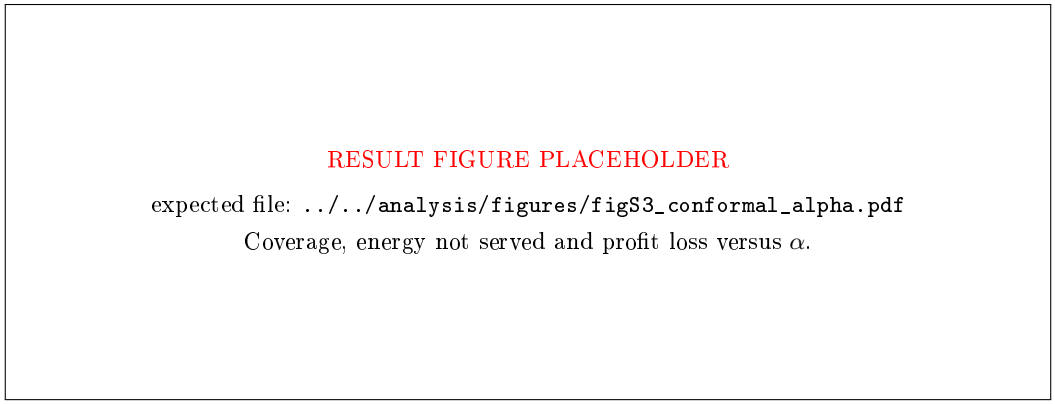


Figure S3: Trade-off of the conformal departure bound.

Table S7: Conformal departure bound: miscoverage level α versus profit and departure shortfall (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

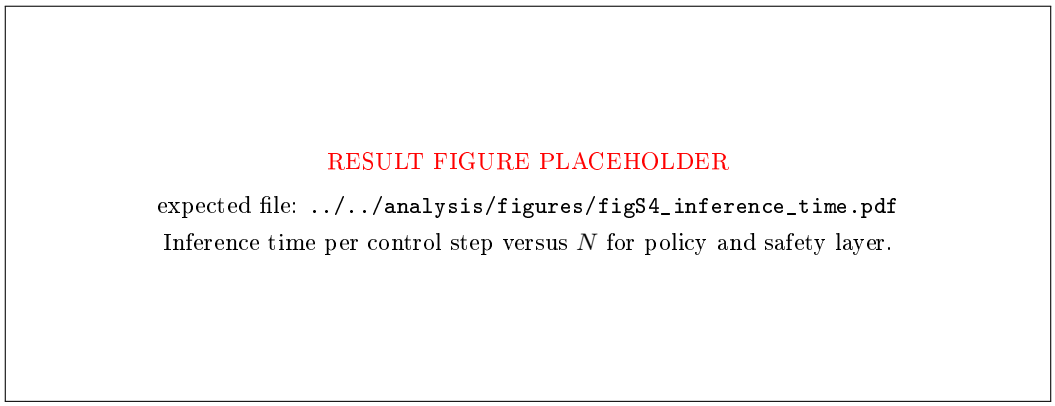


Figure S4: Inference time per control step.

Table S8: Computation: training time per run and inference time per control step (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S9: Optimality gap with respect to $J^{\mathrm{LP,ceil}}$ and J^{LP} per station size N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S10: Sensitivity studies at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW (all three profiles).

Setting	Gap [%]	Unserved [%]	Violation [kWh]	Infeasible
Main setting	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Desired state of charge 0.8	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
$\Delta t = 5$ min	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Penalty weights 1 and 100 €/kWh	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]

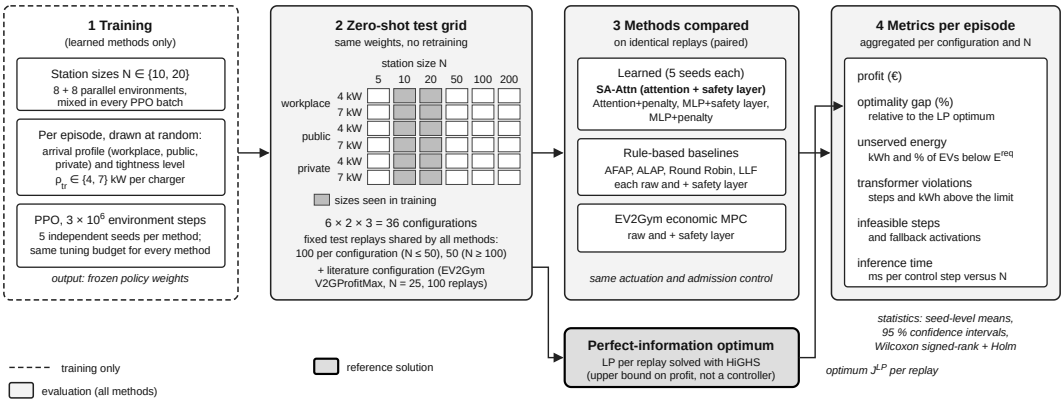


Figure S5: Experimental pipeline. (1) The learned methods train on mixed station sizes, arrival profiles and ratings. (2) A zero-shot test grid has 36 configurations plus the literature configuration, with sizes seen in training shaded. (3) All methods run on identical replays. (4) Metrics are computed per episode, and the perfect-information LP provides the reference optimum of every replay.

RESULT FIGURE PLACEHOLDER

expected file: `../../analysis/figures/fig5_example_day.pdf`

Aggregate power, $C(t)$, $-C_{\mathrm{rev}}(t)$, the band between $\sum_i l_i$ and $\sum_i u_i$, and the price over one episode; fallback steps marked.

Figure S6: Example episode of SA-Attn (workplace profile, $N = 20$, tight rating).